MIL-F-16377/57A(SH)
9 May 1983
SUPERSEDING
MIL-F-16377/57(SHIPS)
23 August 1972

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, 60, 80 AND 120 WATTS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 348, 349 AND 350

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Department and Agencies of the Department of Defense.

(A) The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

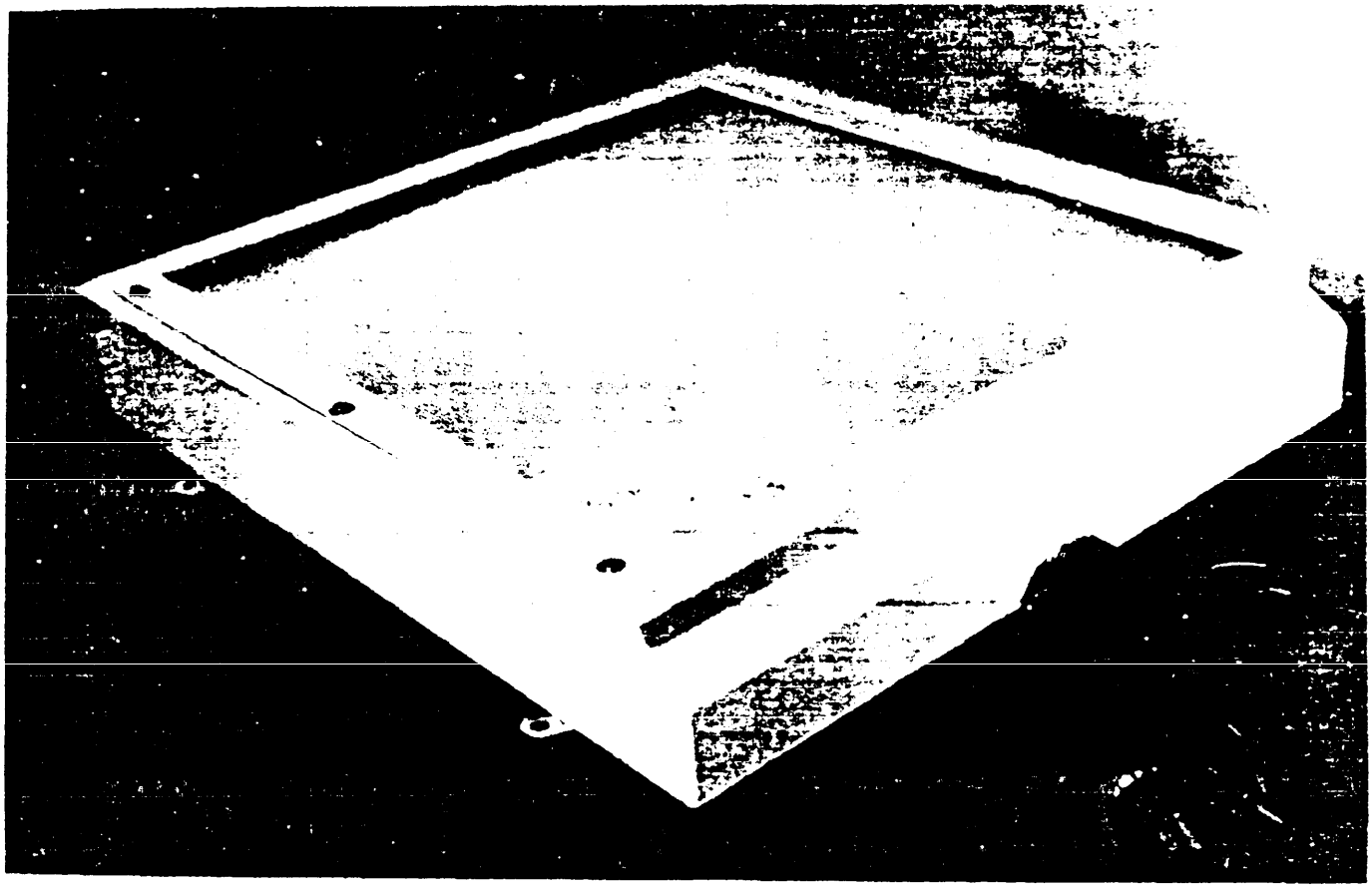


FIGURE 1. Typical view of Symbols 349 and 350.

All dimensions in inches.
For tolerances other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(A) Denotes changes

FSC 6710

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:51Z
Check the source to verify that this is the current version before use.

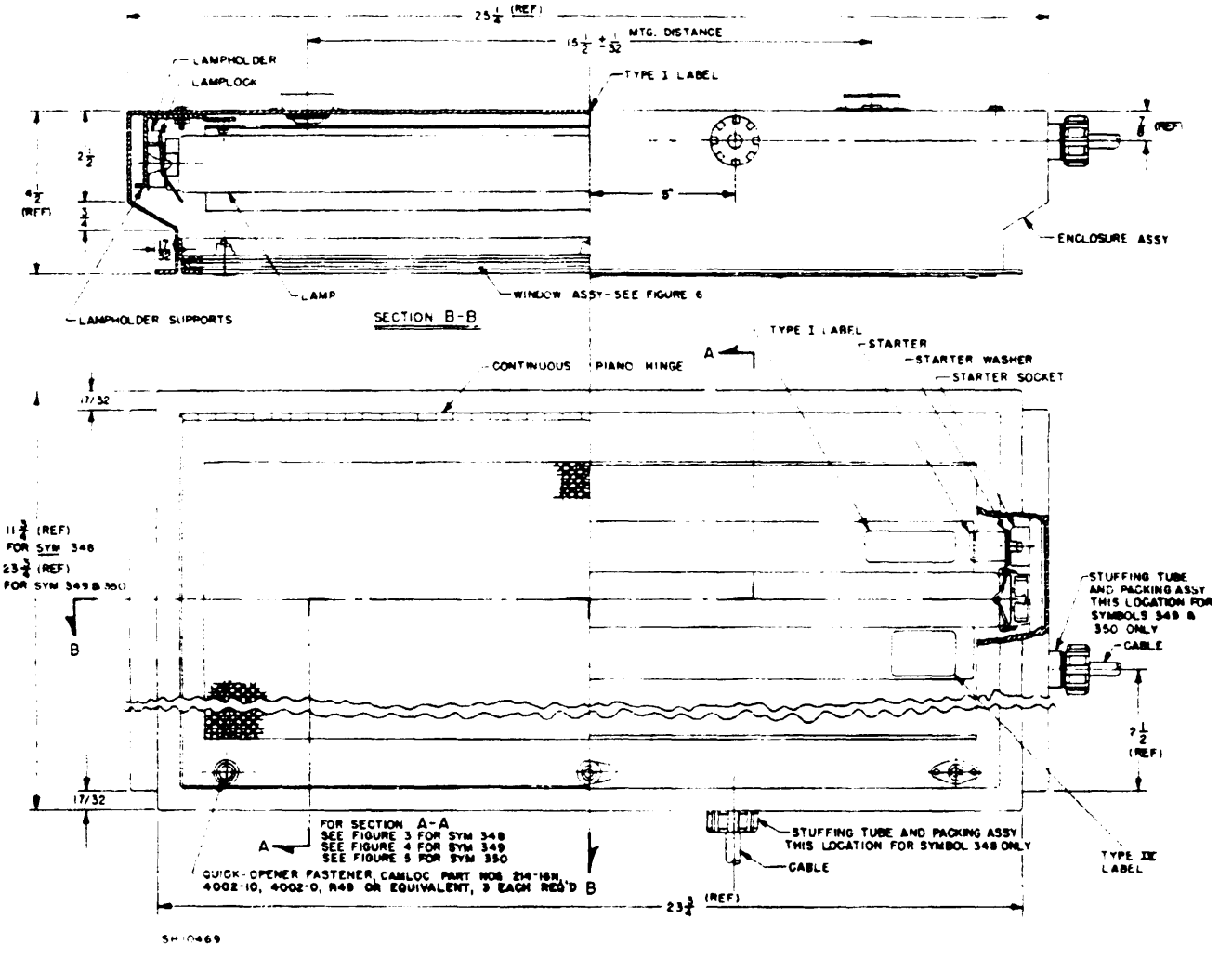
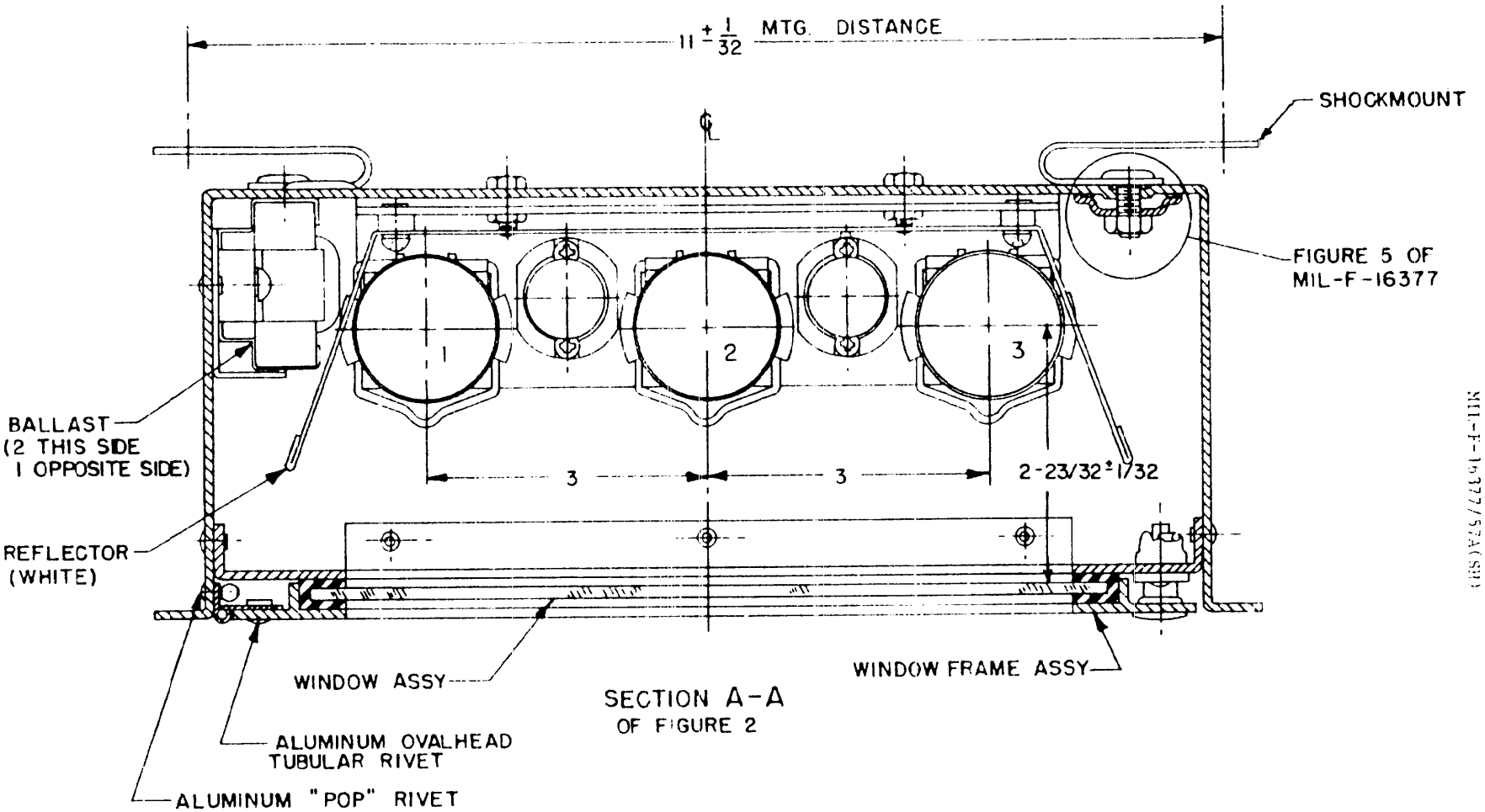


FIGURE 2. Typical views of symbols 348, 349 and 350.

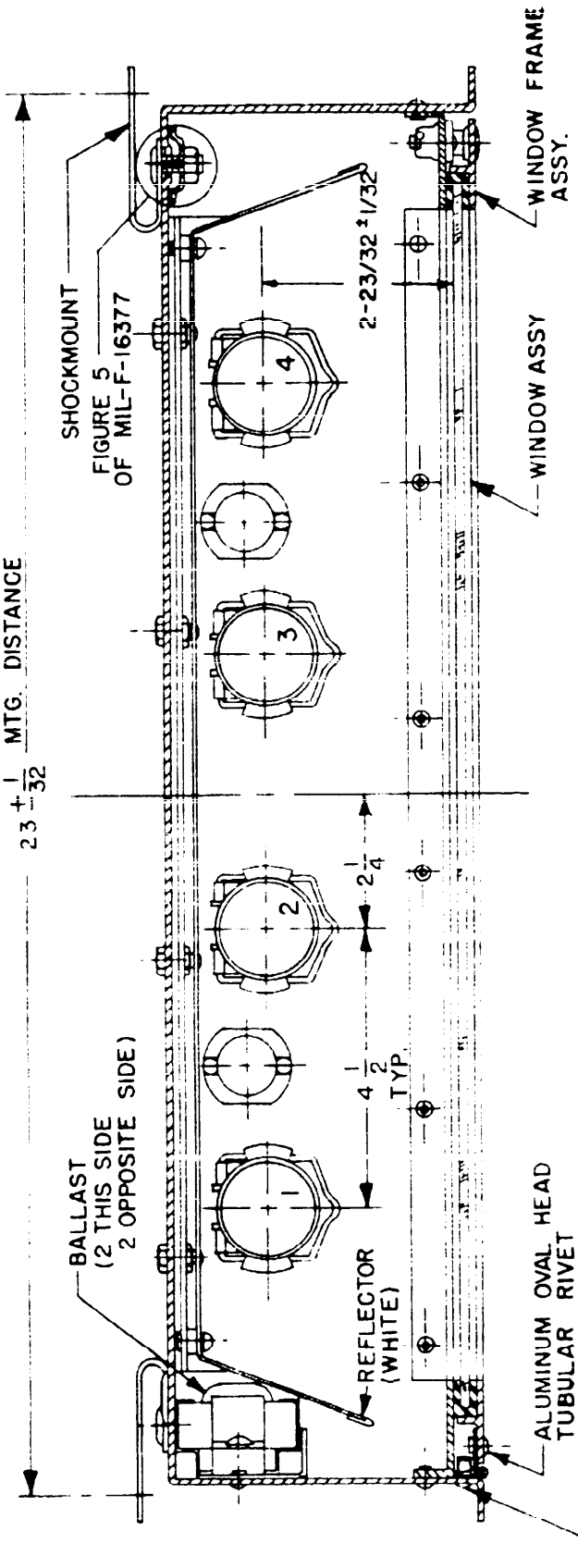
Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the current version before use.



NOTE: For red and white illumination installing activity shall provide a red filter tube on #2 lamp.

FIGURE 3. Typical section view of symbol 348.

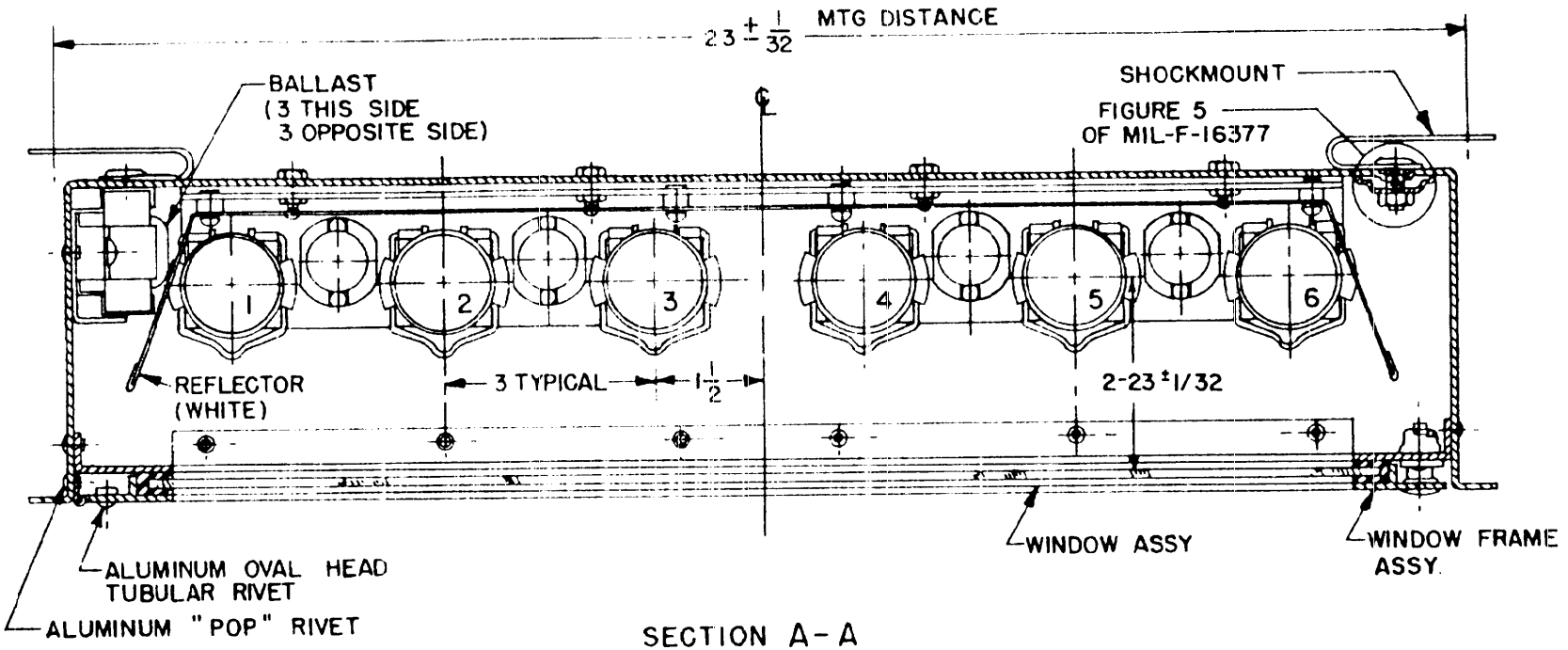
Source: https://assist.dla.mil -- Downloaded: 2026-01-18T08:51Z
Check the source to verify that this is the current version before use.



NOTE: For red and white illumination installing activity shall provide a red filter tube on #2 lamp.

FIGURE 4. Typical section of symbol 349.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the currentversion before use.



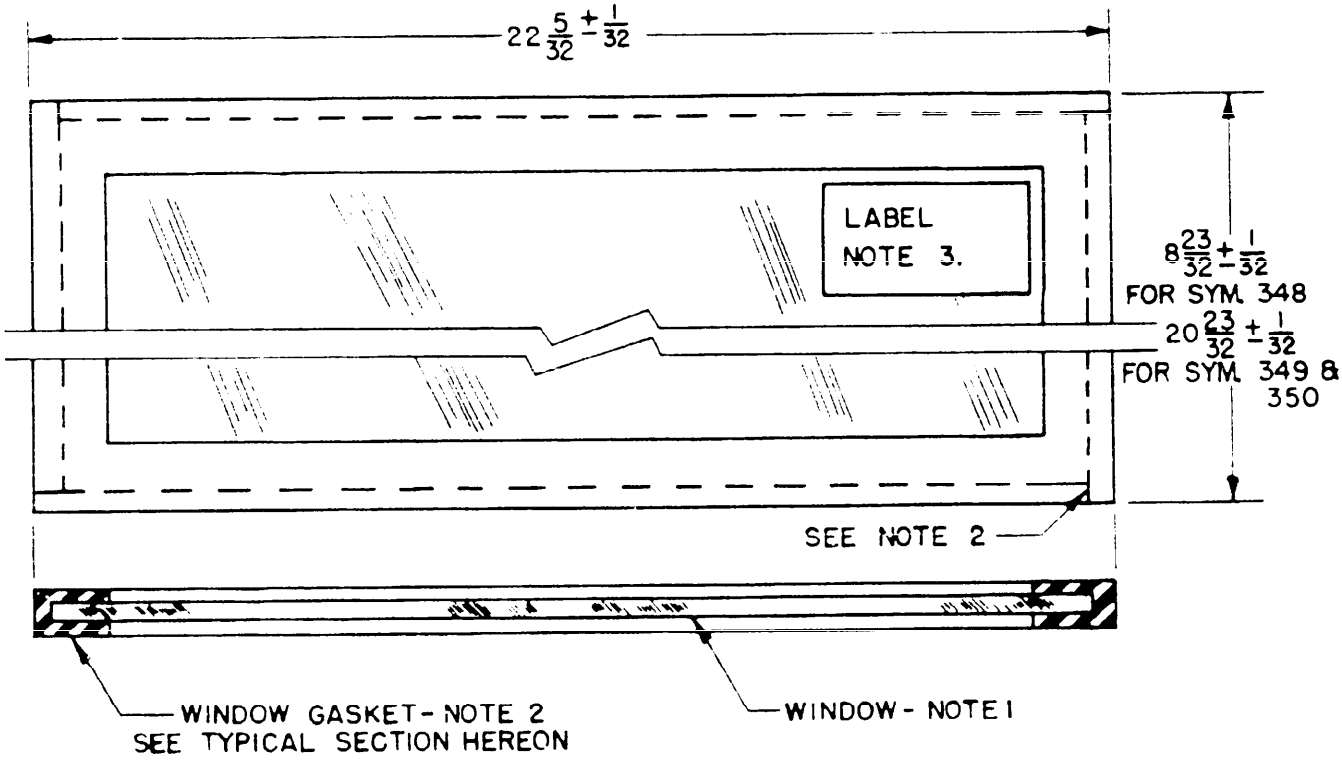
SECTION A-A
OF FIGURE 2

SH 10472

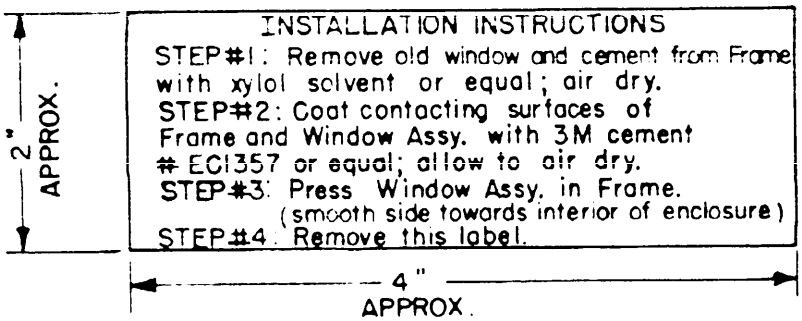
NOTE: For red and white illumination installing activity shall provide red filter tubes on #2 and #5 lamps.

FIGURE 5. Typical section of symbol 350.

Source: https://assist.dla.mil -- Downloaded: 2026-01-18T08:51Z
Check the source to verify that this is the current version before use.

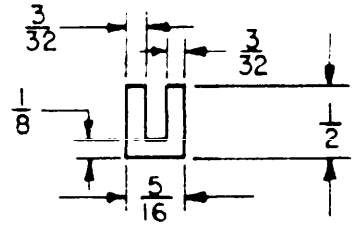


(a) Window Assembly



SH 10473

(b) LABEL - Note 3.



STANDARD RMA TOLERANCES FOR EXTRUSION SHALL APPLY

(c) GASKET - Typical Section

NOTES:

1. Material for window shall be 1/8 inch thick, clear, prismatic (one side only), heat resistant, cast acrylic plastic sheet, type CLEAR-2 of MIL-P-24191.
2. Material for window gasket shall be neoprene rubber 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.
3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-F-16377. Location and the information to be shown on this label shall be as shown hereon.

FIGURE 6. Window assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the current version before use.

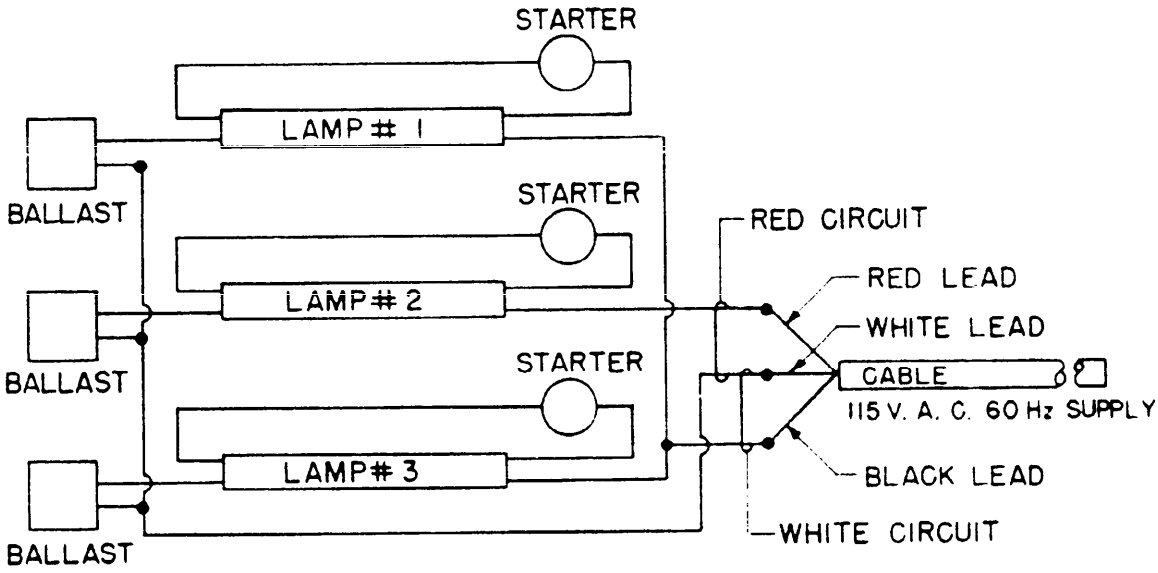


FIGURE 7. Wiring diagram for Symbol 348.

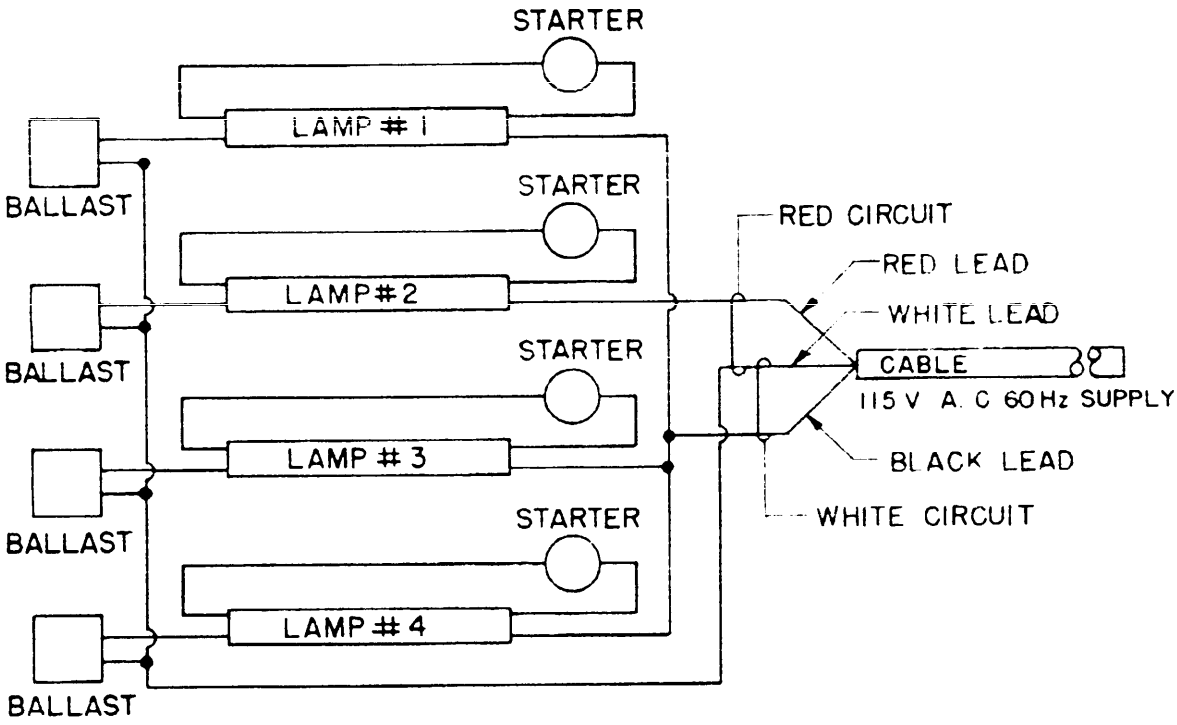


FIGURE 8. Wiring diagram for symbol 349.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the currentversion before use.

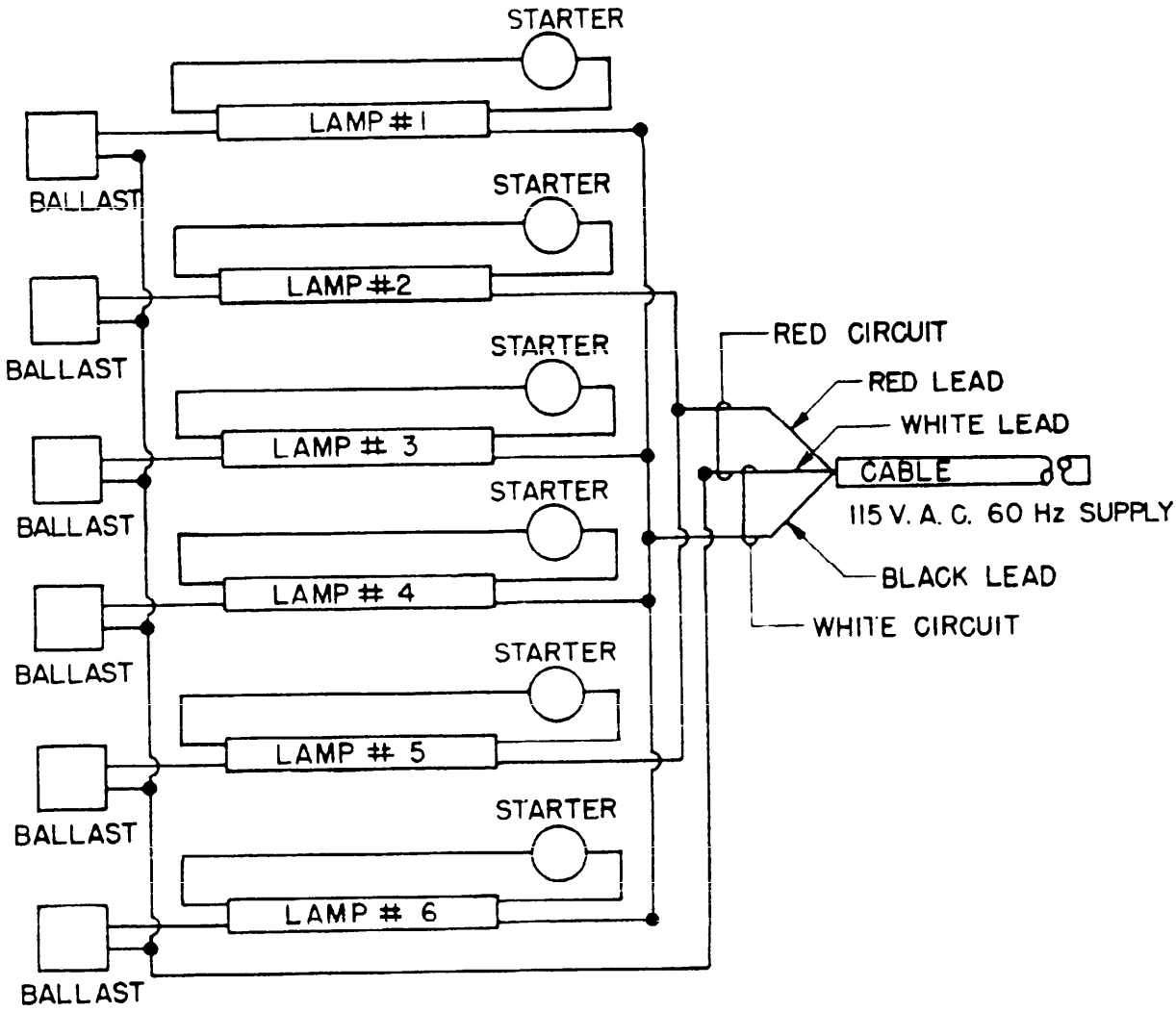
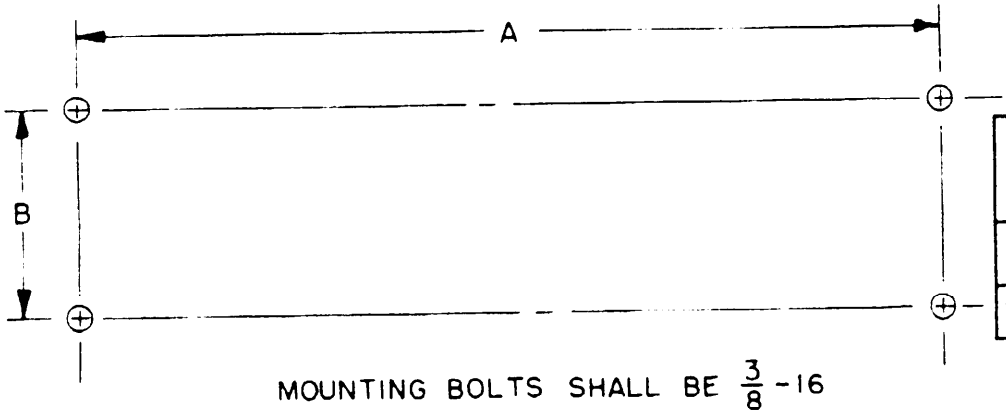


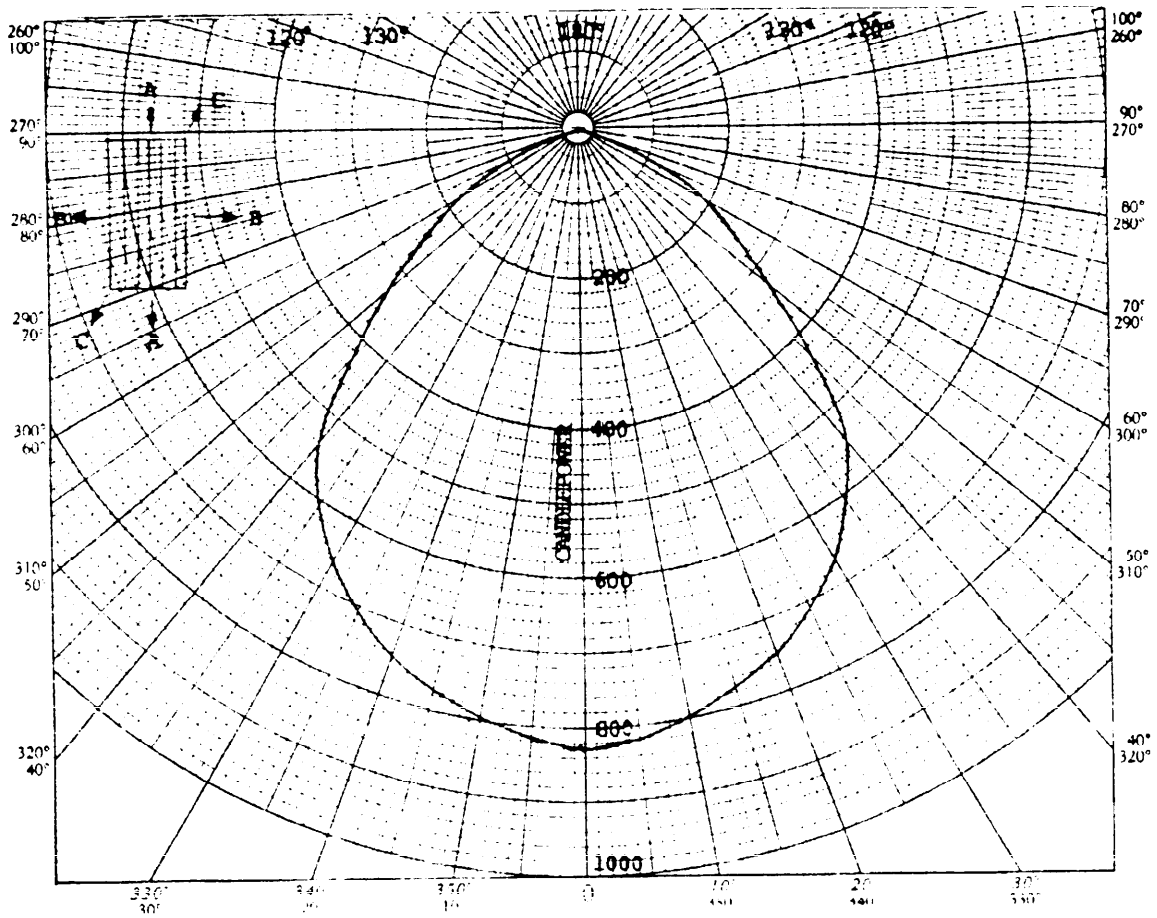
FIGURE 9. Wiring diagram for Symbol 350.



SYMBOL	DIMENSION	
	A	B
348	11	15 1/2
349,350	23	15 1/2

FIGURE 10. Mounting dimensions.

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the currentversion before use.



ANGLE	CANDLEPOWER* SYMBOL 348
0	825
5	810
10	800
15	760
20	750
25	715
30	670
35	620
40	550
45	440
50	330
55	240
60	180

ANGLE	CANDLEPOWER* SYMBOL 348
65	130
70	110
75	90
80	45
85	20
90	10
95	5
100	0
105	0
110	0
115	0
120	0
125	0

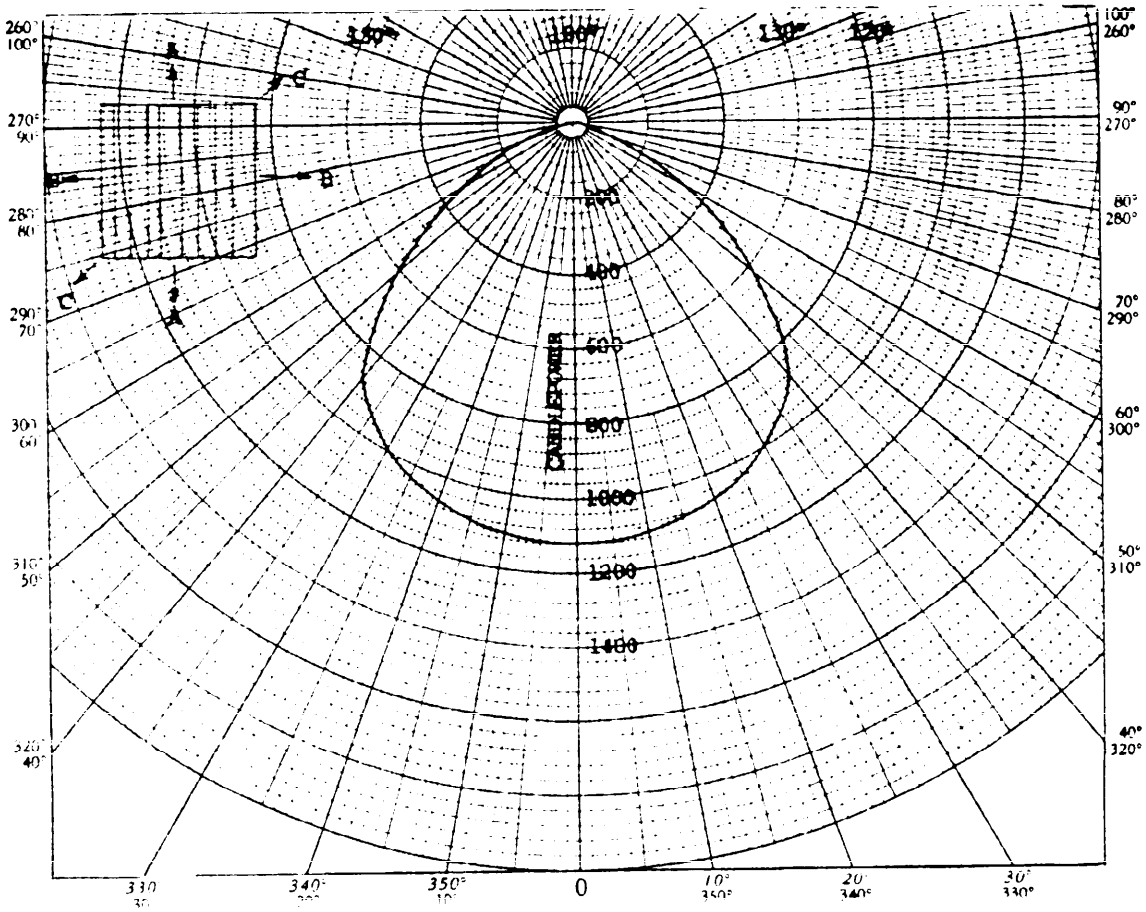
*Average values for planes A-A, B-B and C-C.

Photometric test data:

Distance - - - 10 feet.
Plane - - - - At right angles to long axis of fixture through geometric center of lamp.
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 1280 lumens per lamp.

FIGURE 11. Candlepower distribution curves for Symbol 348.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:51Z
Check the source to verify that this is the current version before use.



ANGLE	CANDLEPOWER* SYMBOL 349
0	1120
5	1120
10	1100
15	1080
20	1070
25	1035
30	1000
35	935
40	890
45	825
50	600
55	450
60	270

ANGLE	CANDLEPOWER* SYMBOL 349
65	220
70	165
75	55
80	45
85	25
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

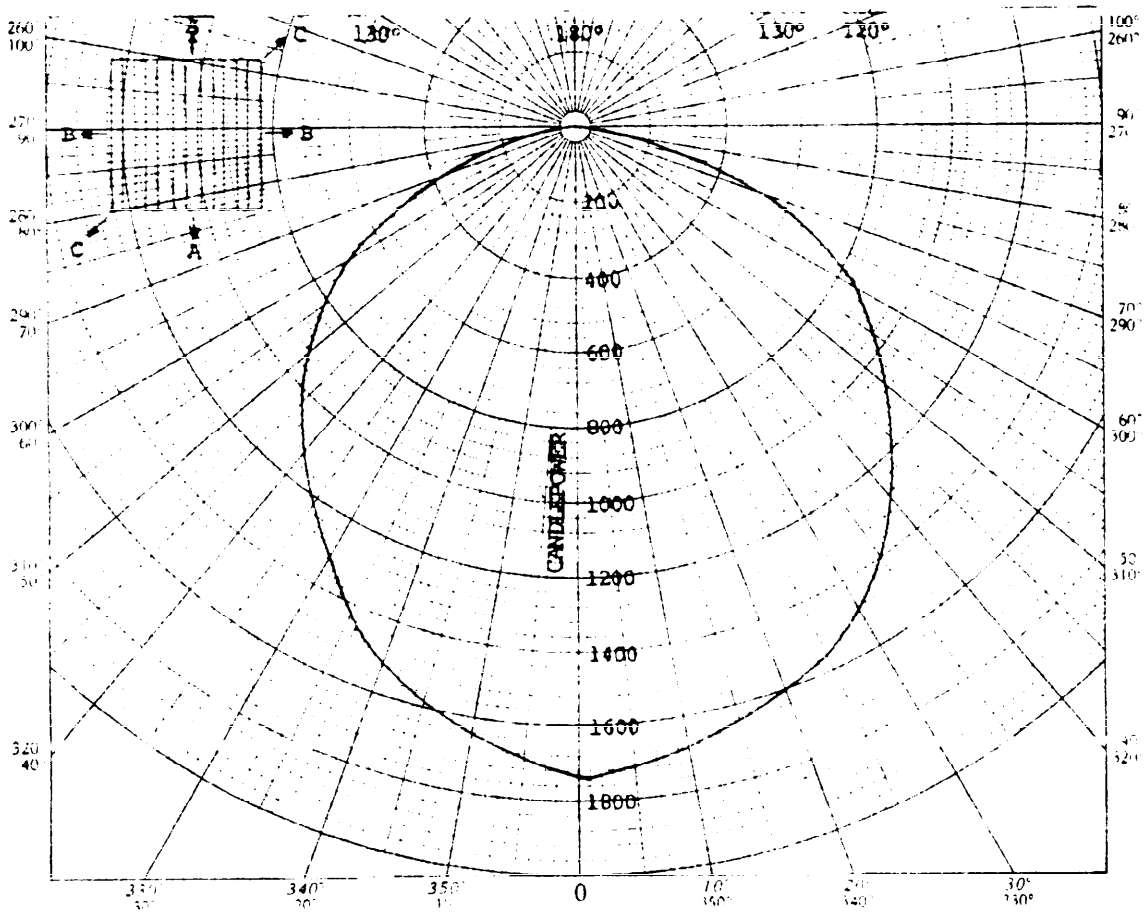
*Average values for planes A-A, B-B and C-C.

Photometric test data:

Distance - - - 10 feet.
Plane - - - - At right angles to long axis of fixture through geometric center of lamp.
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 1280 lumens per lamp.

FIGURE 12. Candlepower distribution curves for Symbol 349.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:51Z
Check the source to verify that this is the current version before use.



ANGLE	CANDLEPOWER* SYMBOL 350
0	1760
5	1760
10	1740
15	1720
20	1670
25	1630
30	1550
35	1475
40	1385
45	1250
50	880
55	660
60	440

ANGLE	CANDLEPOWER* SYMBOL 350
65	330
70	220
75	110
80	55
85	25
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

*Average values for planes A-A, B-B and C-C.

Photometric test data:

Distance - - - [illegible] feet.
Plane - - - - At right angles to long axis of [illegible] through center of lamp.
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F.
Lamps: Use lamps furnished with fixtures aged [illegible] for at least 40 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 1,800 lumens for lamp.

FIGURE 15. Candlepower distribution curve - [illegible] Symbol 350.

Source: https://assist.dla.mil -- Downloaded: 202640-18T08:51Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Type: I
Class: 2
Symbol 348: Fixture, flush mounting, with three 20-watt cool white fluorescent lamps and clear prismatic window.
Symbol 349: Fixture, flush mounting, with four 20-watt cool white fluorescent lamps and clear prismatic window.
Symbol 350: Fixture, flush mounting, with six 20-watt cool white fluorescent lamps and clear prismatic window.

Design:

(A) Electrical rating: 120 volts, 60 Hertz, 60, 80, and 120 watts.
Dimensions and configuration: See figures 1 through 5.
Distribution curves: Candlepower - see figures 11, 12, and 13.

(A) Cable: 10 feet of type SJO or SJT, construction in accordance with J-C-580 except size AWG 16, 3 conductor (black, white, red) in accordance with J-C-580.

Wiring diagram: See figures 7, 8, and 9.
Window assembly: See figure 6.
Mounting dimensions: See figure 10.
Enclosure: Dripproof 15 degrees.
Weight:
Symbol 348: 18 pounds maximum
Symbol 349: 24 pounds maximum
Symbol 350: 27 pounds maximum
Labels: Labels shall be furnished in accordance with the following types of MIL-F-16377:
(a) Type I
(b) Type IV - One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I by name and stock number only.
Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

TABLE I. Standard parts.

(A) Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	MS15589-5	6240-00-152-2996	Industry Number F20T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	----------
Lampholder	MIL-L-970/11	6250-00-284-0481	----------
Startersocket	MIL-L-970/13	6250-00-295-2738	----------
(A) Starter	8W, type III of W-S-739	6250-00-299-2884	Commercial Designation: FS2
Ballast	M16377/44-001	6250-00-569-9502	----------
Shockmount: Symbol 348	M16377/41-009	6210-00-973-1747	See 3.5.16 of MIL-F-16377
Shockmount: Symbols 349, 350	M16377/42-008		
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished See 3.7.7 of MIL-F-16377

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the currentversion before use.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II
Window assemblies:	See table III
Quality conformance:	
Comparison inspection:	Same as the first article inspection
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

TABLE II. First article and comparison inspection for complete fixtures.

Part	Requirement paragraph of MIL-F-16377	Test paragraph of MIL-F-16377	Remarks
Examination	3.13	4.5	-------
Operation	3.13	4.8.1	-------
Dielectric withstanding voltage	3.6.15	4.8.2	-------
Insulation resistance	3.6.16	4.8.3	-------
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof 15 degrees
Shock	3.5.12	4.8.7	-------
Vibration	3.5.13	4.8.8	-------
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof 15 degrees
Dielectric withstanding voltage	3.6.15	4.8.2	-------
Insulation resistance	3.6.16	4.8.3	-------
Noise	3.5.14	4.8.9	-------
Electromagnetic interference	3.6.14	4.8.16	-------
Continuity of grounding	3.6.13.4	4.8.12	-------
Optical uniformity	3.7.4.1.1	4.8.4	-------
Light output	3.7.1	4.8.6	Candlepower
Magnetic permeability	3.4.8	4.8.18	-------

TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph of MIL-F-16377	Test paragraph of MIL-F-16377	Remarks
Examination	3.13	4.5	-------
Optical uniformity	3.7.4.1.1	4.8.4	-------

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:51Z
Check the source to verify that this is the current version before use.

General information: Assignment of Military part numbers and general information is shown in table IV.

TABLE IV. Part number and general information.

Military Part number M16377/57-	Item name	National Stock Number (NSN)	Remarks
348	Complete fixture, Symbol 348	Not assigned	-------
349	Complete fixture, Symbol 349	Not assigned	-------
350	Complete fixture, Symbol 350	Not assigned	-------
001	Window assembly for Symbol 348	----------	Figure 6
002	Window assembly for Symbols 349 and 350	----------	Figure 6

Preparing activity:
Navy - SH
(Project 6210-N 588)

Source: https://assist.dla.mil -- Downloaded: 202610-18T08:51Z
Check the source to verify that this is the current version before use.